«interface»
command::Command

+ *execute(): boolean*
+ *getRequiredEnergy(): double*

Responsibilities
-- Interface for executable robot command operations.

«entity»
model::ReconDrone

- flightAltitude: double

+ executeScanPattern(bounds: Rectangle): void
+ returnToBase(): void

Responsibilities
-- Aerial vehicle subclass for rapid reconnaissance.

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«command»
command::ScanAreaCommand

- drone: ReconDrone
- targetBounds: Rectangle

+ execute(): boolean
+ getRequiredEnergy(): double

Responsibilities
-- Encapsulates aerial reconnaissance flight instructions.

«entity»
model::TrackedRover

- payloadCapacity: double
- attachedRobodog: Robodog

+ traversePath(path: List<Point>): void
+ deploySubUnit(): Robodog

Responsibilities
-- Ground vehicle carrying deployable quadruped subunits.

«abstract»
model::Robot

robotId: String
batteryLevel: double
currentPosition: Point

+ moveTo(destination: Point): boolean
+ consumeBattery(amount: double): void
+ *executeTask(): boolean*

Responsibilities
-- Abstract base class for physical robotic hardware.

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«entity»
model::Robodog

- searchRadius: double

+ performRadiusSweep
(center: Point, radius: double)
: List<SurvivorEntity>

Responsibilities
-- Deployable quadruped subunit for localized perimeter scans.

«interface»
strategy::SearchStrategy

/+ computePath(bounds: Rectangle)
: List<Point>/

Responsibilities
-- Common interface for grid coverage algorithms.

«strategy»
strategy::LawnmowerScanStrategy

+ computePath(bounds: Rectangle): List<Point>

Responsibilities
-- Calculates systematic area grid coverage patterns.

«entity»
model::RescueHelicopter

- maxLiftCapacity: int
- rotorSpeed: double

+ flyToLandingZone(lz: Point): boolean
+ extractTarget(survivor: SurvivorEntity): void

Responsibilities
-- Heavy-lift extraction flight asset for recovering survivors.